

# Tahsin Raihan Robbani

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🐙 GitHub | 🔗 LinkedIn | 📁 Portfolio

## Research Interests

Medical Image Analysis, Computer Vision, Deep Learning, Large Language Models (LLMs), Artificial Intelligence, Bioinformatics, Human-Computer Interaction (HCI), and Scalable IT Systems.

## Education

### BRAC University

Dhaka, Bangladesh

*B.Sc. in Computer Science and Engineering* — **CGPA: 3.83 / 4.00**

Sep 2022 – Sep 2026

- **Relevant Coursework:** Artificial Intelligence, Algorithms, Operating Systems, Automata & Computability, Compiler Design, Probability & Stochastic Processes, Discrete Mathematics, Bioinformatics

## Research Experience

### Neuro-Symbolic AI for Optical Biopsy-Level Polyp Diagnosis in Endoscopic Video

2025 – 2026

*Undergraduate Thesis, BRAC University CSE* | Status: Manuscript in preparation (Target Venue: MICCAI 2027)

**Supervisors:** Prof. Dr. Swakkhar Shatabda (Supervisor), Prof. Dr. Md. Golam Rabiul Alam (Co-supervisor)

- Developed a real-time optical biopsy pipeline for colonoscopy video, combining an ensemble detector (YOLOv8, RT-DETR, MedSAM2) with contrastive self-supervised ViT representations and 60 extracted visual biomarkers.
- Built an interpretable Mixture-of-Experts model with calibrated confidence that encodes clinical NICE/ESGE guidelines, predicting not only polyp's classification but also calibrated cancer risk scores to help endoscopists make immediate, evidence-backed resection decisions.
- Evaluated end to end on 349 annotated colonoscopy videos across 16 pathology classes without test-time fine-tuning: achieved high video-level detection sensitivity, 91.98% polyp classification accuracy, and 93.2% sensitivity on high-risk adenomas.

## Academic Experience

### Student Tutor (ST) — Discrete Mathematics (CSE230)

Mar 2026 – Present

*Department of Computer Science and Engineering, BRAC University*

Dhaka, Bangladesh

- Appointed for Spring and Summer 2026; mentor undergraduate students through formal logic proofs, set theory, induction, predicate calculus, and graph algorithms during weekly consultation hours.
- Coordinate regularly with course faculty to align consultation topics with lecture progress, assist in lab sessions, and address common student difficulties.

## Selected Projects

### Crash-Consistent Metadata Journaling for VSFS | C, Systems Programming, POSIX

2024

- Engineered a write-ahead metadata journaling layer for a virtual file system (VSFS) in C to guarantee crash consistency and transactional durability across sudden system interruptions.
- Implemented atomic commit protocols and redo-log recovery routines, eliminating inode table and block bitmap corruption under simulated disk crash fault injection.

### Oi Mama: A Banglish Domain-Specific Language Compiler | Python, Lexer/Parser, AST, PyPI

2026

- Designed and published an open-source domain-specific programming language compiler on PyPI featuring custom lexical analysis and Python Abstract Syntax Tree (AST) code generation.
- Built language runtime support for object-oriented paradigms, asynchronous routines, and a tailored compiler diagnostic engine providing localized error reporting.

### Gesture Board: Touchless Real-Time Collaborative Canvas | React, MediaPipe, Supabase

2026

- Engineered a low-latency collaborative digital whiteboard utilizing MediaPipe hand tracking for continuous spatial coordinate mapping, decoupling inference from canvas rendering over Supabase Realtime.

## Technical Skills

**Languages:** Python, C, C++, Java, JavaScript, TypeScript, SQL, R

**Machine Learning & Vision:** PyTorch, Tensorflow, YOLOv8, RT-DETR, MedSAM2, U-Net++, SimCLR, MediaPipe, OpenCV, Calibration

**Systems & Architecture:** POSIX Systems Programming, Virtual File Systems, Lexer/Parser Design, Git, Linux

**Tools & Full-Stack:** Node.js, PostgreSQL, MongoDB, Supabase, React, Svelte, LaTeX, Arduino